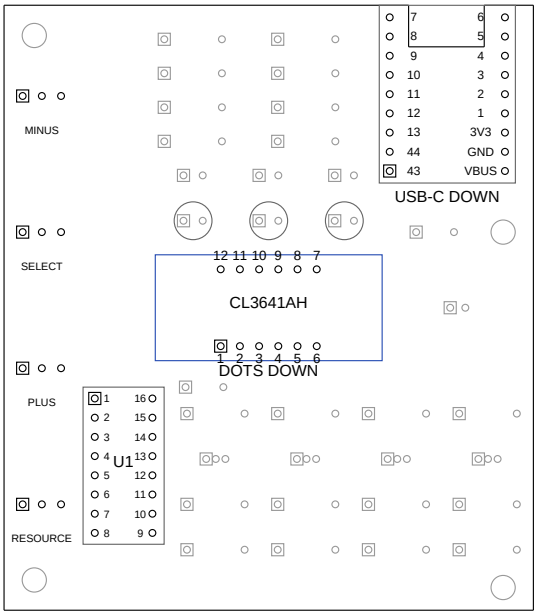
# COMPACT COUNTER: actual-size fit check

CL3641AH + Waveshare ESP32-S3-Zero | revision E1 | 70 x 80 mm

Print at 100%. Disable Fit to Page. Verify the 100 mm calibration line.

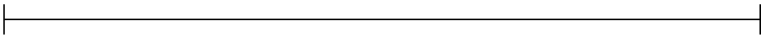



J1: Waveshare S3-Zero
18 x 23.5 mm body.
Two 9-pin headers.
2.54 mm pitch; 15.24 mm rows.
Antenna UP over the notch.
USB-C DOWN toward display.

DS1: CL3641AH
30 x 14 mm body.
Six pins per row.
2.54 mm pitch; 10.16 mm rows.
Decimal points DOWN.

U1: SN74HC595N DIP16
Notch UP; pin1 upper-left.

SW: square pad = signal.
Choose ONE round GND pad.
2.54 or 5.08 mm spacing.
Wire buttons if needed.



THIS LINE MUST BE 100 mm

1. Compare the unplugged S3-Zero and display against their hole rows.
2. Check every module pin label. The original 30-pin ESP32 does not fit J1.
3. Check button lead spacing. Use wires if the unknown switch body does not fit.
4. Seat the module on female headers so its underside clears the carrier.
5. Keep the ceramic antenna and the notch clear of metal and nearby components.
6. This sheet checks nominal geometry only. Physical fit and operation are untested.

The breadboard model is separate: it uses your normal ESP32 and original firmware.
Assembly, BOM, GPIO map and build instructions: compact-cl3641ah/README.txt.